

Adult Income Prediction — Day 2 Summary

Preprocessing, first supervised models, and evaluation — UCI Adult (Census Income) dataset

1. Preprocessing Choices

Numeric features: age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week — median imputation (robust to the heavily skewed capital-gain/capital-loss distributions) then StandardScaler (keeps features on a comparable scale for the linear model).

Categorical features: workclass, education, marital-status, occupation, relationship, race, sex, native-country — most-frequent imputation then OneHotEncoder(handle_unknown='ignore') to avoid imposing false ordinal structure and to safely handle any unseen category at inference time. Alternatives considered and skipped: mean imputation (too sensitive to outliers/skew), ordinal encoding (inappropriate for nominal categories), and target encoding (risks label leakage). Both pipelines are wrapped in a single ColumnTransformer and cloned per model, so each estimator gets its own preprocessing state (avoids a shared-instance bug across pipelines).

2. Models & Splits

Same stratified 80/20 hold-out test split as Day 1 (random_state=42). Two full pipelines (preprocessing + estimator) were fit on the training set only, never touching the test set until evaluation: Logistic Regression (solver='liblinear', random_state=42) and Decision Tree (random_state=42, unconstrained).

3. Model Comparison (held-out test set)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day 1 baseline (rule)	0.782	0.625	0.225	0.331	0.591	0.326
Logistic Regression	0.851	0.732	0.594	0.656	0.904	0.760
Decision Tree (unconstrained)	0.810	0.601	0.616	0.608	0.743	0.462

Both supervised models beat the Day 1 rule baseline on every metric. Logistic Regression leads on precision (our chosen metric), ROC AUC, and PR AUC by a wide margin.

4. Error Types & Interpretability

Errors: Logistic Regression leans false-negative-heavy (949 FN vs 509 FP — misses more true high earners than it wrongly flags). The Decision Tree is roughly balanced (955 FP vs 899 FN). Since false positives are the costly error for our outreach goal, Logistic Regression's much higher precision (0.732 vs 0.601) still makes it the safer choice despite its FN lean.

Logistic Regression coefficients: capital-gain, being married, education-num, and Exec-managerial occupation push predictions toward >50K; Priv-house-serv occupation, never-married status, being recorded as an own-child/other-relative, and being recorded as female push predictions toward ≤50K (the last reflecting a real historical income gap in this census data, not a causal claim). Several of the largest-magnitude coefficients are specific native-country categories with very few records each, so those should be treated as noisy rather than reliable signals.

Decision Tree — overfitting flag: depth 59, train score 0.9999 vs test score 0.810 — a ~0.19 gap that is a clear overfitting signature; the tree has essentially memorized the training set. Reinforcing this, fnlwgt (a Census sampling weight with no real causal link to income) is the tree's 2nd most important feature by aggregate importance — a sign it is fitting noise rather than genuine structure. The tree's actual top splits (marital-status, then capital-gain, then education-num) are sensible and line up with the logistic regression coefficients, but the excessive depth means the tree needs constraints before it is a fair comparison.

5. Model Selection for Day 3

Continuing with Logistic Regression as the primary model. It wins on every metric (precision 0.732 vs 0.601, ROC AUC 0.904 vs 0.743, PR AUC 0.760 vs 0.462) and is directly interpretable via coefficients, which matters for stakeholder buy-in on an outreach-targeting decision. The Decision Tree is kept as a comparison candidate for capturing feature interactions a linear model misses, but only after being properly regularized — its current numbers are depressed by severe overfitting, not necessarily a real ceiling on tree-based performance.

Planned changes for tomorrow: (1) constrain the Decision Tree (max_depth / min_samples_leaf / ccp_alpha pruning) to fix today's overfitting before re-comparing it; (2) test dropping fnlwgt from the feature set, since it has no causal link to income and is already showing up as a top tree feature; (3) try an explicit 'Missing' category for categoricals instead of most-frequent imputation; (4) evaluate Random Forest / Gradient Boosting as stronger non-linear candidates now that a single unconstrained tree is known to overfit badly on this data.